

Slide 1

Question

1. What is a conditional statement in programming?

A conditional statement in programming is a decision-making statement that executes different blocks of code depending on whether a condition is true or false.

Definition

A conditional statement checks a condition and performs an action based on the result.

Example

If a student's marks are greater than or equal to 40, the student passes; otherwise, the student fails.

Slide 2

Question

2. What does an if statement do?

It is a decision making statement in c/c++ that checks whether a condition is true or false.

Slide 3

Question

3. What is the purpose of an else block?

The else block is used to execute the block of code when the condition is false.

Slide 4

Question

4. When is an else block executed?

Else block is executed when the condition in the corresponding if statement is false.

Slide 5

Question

5. What is a Boolean expression?

A Boolean statement is an expression that evaluates to only two possible values

True

False

Slide 6

Question

6. Write the syntax of a simple if statement

If (condition)

{ //statements to be executed

}.

Slide 7

Question

7. What operator is commonly used to test equality?

Equality operator (==) is used to test whether two values are equal in C/C++

Slide 8

Question

8. What does the modulus (%) operator return?

Modulus operator returns the remainder when one integer is divided by another .

Slide 9

Question

9. How can you check if a number is even?

A number will be even if it leaves a remainder of 0 when divided by 2.

Slide 10

Question

10. How can you check if a number is odd?

A number is said to be odd if it leaves a remainder when divided by 2.

Slide 11

Question

11. What is the purpose of else-if?

The main purpose is to check

Multiple conditions one after another.

Slide 12

Question

12. What is a nested if statement?

A nested if statement is an if statement placed inside another if statement .

Slide 13

Question

13. Can an if statement exist without an else?

Yes if statement can exist without an else statement.

Slide 14

Question

14. What is the output when a false condition is tested in an if statement without else?

When a false condition is tested in an if statement without an else block, the statements inside the if block are not executed. The program simply skips the if block and continues with the next statement.

Slide 15

Question

15. What is the role of braces {} in C/C++ conditionals?

Braces {} are used to group multiple statements into a single block in conditional statements such as if, else, and else if.

Slide 16

Question

16. What does > mean?

The > (greater than) operator is a relational operator used to compare two values.

Slide 17

Question

17. What does < mean?

The < (less than) operator is a relational operator used to compare two values.

Slide 18

Question

18. What does $\geq$ mean?

The $\geq$ (greater than or equal to) operator is a relational operator used to compare two values.

Slide 19

Question

19. What does $\leq$ mean?

The $\leq$ (less than or equal to) operator is a relational operator used to compare two values.

Slide 20

Question

20. What does != mean?

The != (not equal to) operator is a relational operator used to compare two values.

Slide 21

Question

21. What is a switch statement?

A switch statement is a decision-making statement that allows a program to choose one block of code from multiple alternatives based on the value of an expression.

Slide 22

Question

22. What is a case label in a switch statement?

A case label is a value inside a switch statement that specifies a particular option to be matched with the value of the switch expression.

Slide 23

Question

23. Why is break used in switch?

The break statement is used to terminate the execution of a switch case and exit the switch block.

Slide 24

Question

24. What happens if break is omitted?

If the break statement is omitted, the program continues executing the statements of the next case(s) even if their labels do not match. This is called fall-through.

Slide 25

Question

25. What is the default case in switch?

The default case is executed when none of the case labels match the value of the switch expression.

Slide 26

Question

26. Can switch compare strings in C++?

No, a switch statement cannot directly compare strings in standard C++.

A switch statement works only with:

int

char

enum

Other integral types

Slide 27

Question

27. Which statement is better for many constant choices: if-else or switch?

For many constant choices, a switch statement is generally better than an if-else ladder.

Slide 28

Question

28. How do you test whether a value is positive?
A value is positive if it is greater than zero.

Slide 29

Question

29. How do you test whether a value is negative?
A value is negative if it is less than zero.

Slide 30

Question

30. How do you test whether a value is zero?
A value is zero if it is equal to 0.

Slide 31

Question

31. Write a condition to check if age is at least 18
age >= 18

Slide 32

Question

32. Write a condition to check if marks are above 50.
`marks > 50`

Slide 33

Question

33. What is decision making in programming?

Decision making is the process by which a program chooses between different actions based on whether a condition is true or false.

It is done using conditional statements such as:

if

if-else

else if

switch

Slide 34

Question

34. Can nested if statements have multiple levels?

Yes, nested if statements can have multiple levels. An if statement can be placed inside another if statement, and this nesting can continue as needed.

Slide 35

Question

35. What is logical AND (&&)?

The logical AND operator (&&) is used to combine two or more conditions.

It returns:

True (1) if all conditions are true.

False (0) if any one condition is false.

Slide 36

Question

36. What is logical OR (||)?

The logical OR operator (||) is used to combine two or more conditions.

Slide 37

Question

37. What is logical NOT (!)?

The logical NOT operator (!) is used to reverse the result of a condition.

Slide 38

Question

38. How do you check if a year is divisible by 4?
`year % 4 == 0`

Slide 39

Question

39. What is a leap year?

A leap year is a year that has 366 days instead of 365 days.

Slide 40

Question

40. int a = 10, b = 20;

```
if(a > b)
{
cout << "Largest = " << a;
}
else
{
cout << "Largest = " << b;
}
```

Slide 41

Question

41. How do you find the largest of two numbers?

int a = 10, b = 20;

```
if(a > b)
{
    cout << "Largest = " << a;
}
else
{
    cout << "Largest = " << b;
}
```

Slide 42

Question

42. How do you find the largest of three numbers?

int a = 10, b = 25, c = 15;

```
if(a >= b && a >= c)
{
    cout << "Largest = " << a;
}
else if(b >= a && b >= c)
{
    cout << "Largest = " << b;
}
else
{
    cout << "Largest = " << c;
}
```

Slide 43

Question

43. What is an if-else-if ladder?

An if-else-if ladder is a conditional structure used to check multiple conditions one after another.

If the first condition is true, its block executes.

Otherwise, the next else if condition is checked.

If none of the conditions are true, the else block executes.

Slide 44

Question

44. What is the difference between if and switch?

The if statement is used to test complex conditions and ranges, while the switch statement is used to select one block of code from multiple constant choices. if is more flexible, whereas switch is more organized for many fixed options.

Slide 45

Question

45. Can switch use ranges directly?

No, a switch statement in C/C++ cannot directly use ranges such as 1-10 or 50-100.

Slide 46

Question

46. What is fall-through in switch?

Fall-through occurs when a case in a switch statement does not have a break statement. The program continues executing the next case(s) even if their labels do not match.

Slide 47

Question

47. What data type is usually used for conditions?

The Boolean (bool) data type is usually used for conditions in C++.

A Boolean value can have only two values:

true

false

Slide 48

Question

48. Why are conditional statements important?

Conditional statements are important because they allow a program to make decisions and perform different actions based on different conditions.

Slide 49

Question

49. Give one real-life example of a conditional statement.

A real-life example of a conditional statement is:

If it rains, then I will carry an umbrella.

Condition: It rains.

Action/Result: I will carry an umbrella.

Slide 50

Question

50. Name four conditional structures covered.

The four conditional structures covered in C/C++ are:

Simple if

if-else

if-else-if ladder

switch statement